

Lab 3 Report: Topology and Flow Control

gem5 Garnet Network-on-Chip Simulation

Name: _____
Student ID: _____

gem5 v23.0.0.1 July 2026

Abstract

This laboratory extends Garnet with a 16-node bidirectional Ring topology, custom shortest-path routing, and a wormhole-style flow-control mode for single-flit control traffic. Ring and 4x4 Mesh experiments are swept over 50 injection rates. A second sweep compares one VC of depth one, sixteen VCs of depth one, and one wormhole VC of depth sixteen. The Ring has a low-load latency of 13.158 Ruby cycles and reaches a stable throughput of 0.233756 packets/node/Ruby cycle before the finite-window experiment shows blocking. The two 16-slot alternatives sustain approximately 0.248 packets/node/Ruby cycle across the tested range, while the one-slot baseline stalls at 0.09 injection rate.

Contents

1	Experimental Method	2
2	Task 1: Ring Topology	2
2.1	Implementation	2
2.2	Measured comparison	2
3	Task 2: Wormhole Flow Control	4
3.1	Implementation	4
3.2	Comparison	4
4	Conclusion	5

1 Experimental Method

Experiments use gem5 v23.0.0.1, the NULL ISA, and the `Garnet_standalone` protocol. The global frequency is 2 GHz, so one Ruby tick is 0.5 ns. Each run lasts 10000 Ruby cycles, equivalent to 5000 system cycles because the synthetic traffic generator uses a 1 GHz system clock. Injection rates are specified in packets/node/system-cycle; the corresponding offered load per Ruby cycle is

$$\lambda_{Ruby} = \frac{\lambda_{system}}{2}. \quad (1)$$

For received packet count N_r , the reported throughput is

$$T = \frac{N_r}{N_{nodes} 10000}. \quad (2)$$

Both tasks use vnet 0, which carries single-flit control packets and uniform random destinations. A point is called *stalled* when source injection acceptance is below 0.9 of the expected Bernoulli-generated packet count. This indicates backpressure or a cyclic congestion condition in the finite run; it does not mean that the gem5 process crashed.

2 Task 1: Ring Topology

2.1 Implementation

`configs/topologies/Ring.py` constructs exactly 16 routers, one external link per router, and two directed internal links per router. The two ports are named `Clockwise` and `CounterClockwise`. The custom routing method in `RoutingUnit::outportComputeCustom()` computes clockwise and counter-clockwise modular distances and selects the shorter direction; ties are resolved clockwise. Thus packets use shortest paths on the physical bidirectional ring.

The relevant build command is:

```
scons --ignore-style build/NULL/gem5.opt \\  
  PROTOCOL=Garnet_standalone -j$(nproc)
```

Directed smoke tests verified one-hop paths for 0 to 15 and 15 to 0, an eight-hop antipodal path for 0 to 8, and a five-hop path for 3 to 14.

2.2 Measured comparison

The sweep contains 50 rates from 0.01 through 0.50 for both the Ring and a 4x4 Mesh with the same 16 nodes.

Table 1: Task 1 summary from the measured CSV data.

Topology	Low-load latency	Low-load hops	Max stable throughput	Max stable rate	First stalled rate
16-node Ring	13.158	4.072	0.233756	0.47	0.46
4x4 Mesh	9.936	2.465	0.248331	0.50	not observed

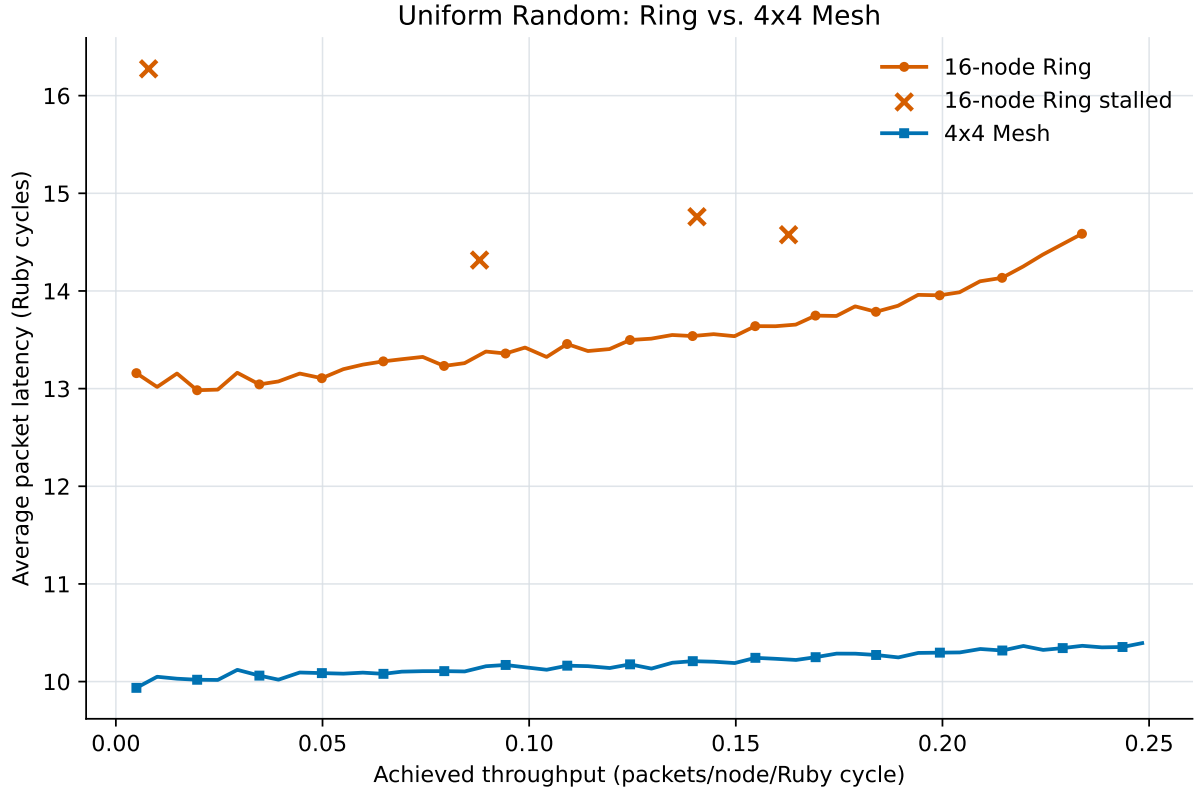


Figure 1: Task 1 latency-throughput comparison. Crosses mark points with source acceptance below 0.9.

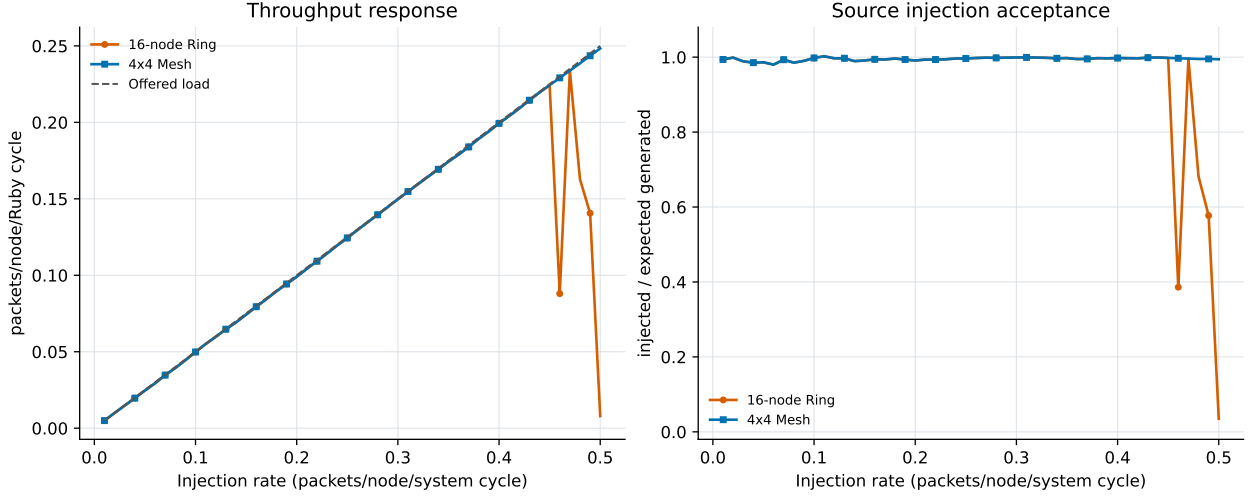


Figure 2: Task 1 throughput and source injection acceptance.

At low load the Ring has higher latency because its average shortest path is about four hops, compared with about 2.5 hops for the Mesh. The Ring also has fewer independent routes and contains a cyclic channel dependency. Therefore, near saturation, acceptance can drop sharply and later recover at an adjacent sample: this is finite-window congestion behavior, not a claim that the Ring has a monotonic throughput curve. The Mesh remains stable through the tested rate of 0.50.

3 Task 2: Wormhole Flow Control

3.1 Implementation

The new `-wormhole` option is represented by the Garnet network parameter `wormhole`. When enabled, control vnet buffers are configured with depth 16. The implementation is intentionally limited to single-flit `HEAD_TAIL_` packets, as required by the lab.

Normally a VC becomes idle after its tail flit and its credit is returned. In wormhole mode, if another single-flit control packet is already queued in the same input VC, the VC remains active, one credit is returned, and the next packet is routed independently. Output VC selection can reuse an active VC when it has downstream credit. The network interface similarly reuses a control VC only when a credit is available. These changes preserve normal packet ownership for all non-wormhole traffic.

3.2 Comparison

The three configurations provide the lab’s requested equivalent 16-slot comparison:

Table 2: Task 2 measured summary.

Configuration	Low-load latency	Max stable throughput	Max stable rate	First stalled rate
VC=1, depth=1	13.365	0.039288	0.08	0.09
VC=16, depth=1	13.158	0.248237	0.50	not observed
VC=1, depth=16 (wormhole)	13.158	0.248250	0.50	not observed

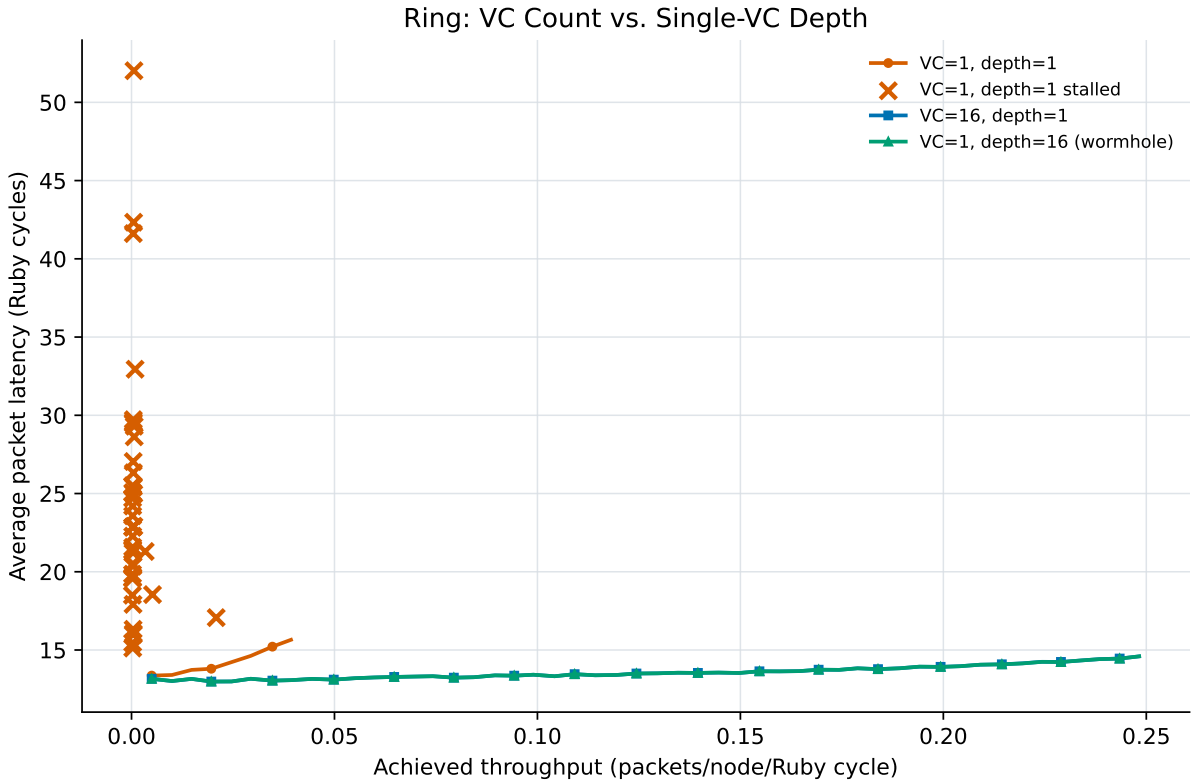


Figure 3: Task 2 latency-throughput comparison.

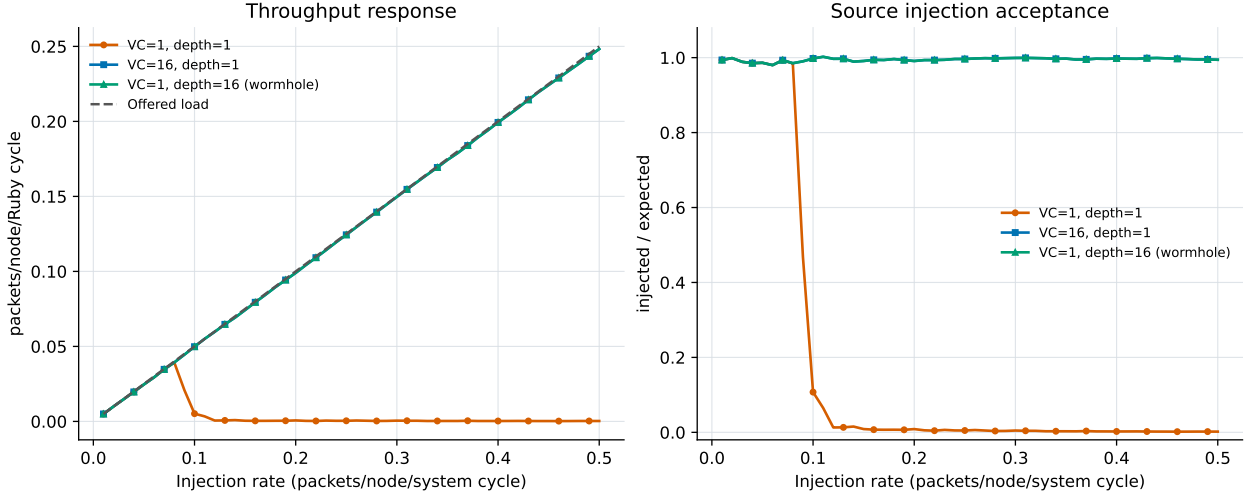


Figure 4: Task 2 throughput and source injection acceptance.

With one slot, a packet must wait for the complete credit round trip before the source can inject the next packet. This creates severe head-of-line blocking, and the $VC=1/\text{depth}=1$ configuration stalls at 0.09. Sixteen VCs provide independent routing and arbitration contexts. A single depth-16 VC does not create more physical bandwidth; instead, it allows up to sixteen single-flit packets to occupy the buffer while credits drain. In this single-flit control workload, its measured throughput is essentially the same as $VC=16/\text{depth}=1$.

The physical link still transmits at most one flit per cycle in every case. The improvement comes from keeping links busy and hiding credit latency, not from increasing link width or link rate. For multi-flit packets, a deeper single VC could still suffer head-of-line blocking, so this result should not be generalized beyond the specified single-flit workload.

4 Conclusion

The implementation satisfies both Task 1 and Task 2. The Ring topology and custom shortest-path routing work for directed and synthetic traffic. Its longer paths and cyclic resource dependencies explain the higher low-load latency and high-load stalls relative to the Mesh. The wormhole mode correctly permits multiple queued single-flit control packets in one VC. Under the tested workload, buffer capacity is the dominant difference: one slot is insufficient, while either sixteen independent VCs or one depth-16 wormhole VC reaches the physical network’s throughput limit.